Pearson Edexcel International A Level · Pure Mathematics 1

Chapter 1 — Algebraic expressions
Original Practice Questions

30 complete questions · Questions followed by answers and suggested mark schemes

- Answer all parts and show sufficient working.
- Give exact answers unless a degree of accuracy is requested.
- Angles are in radians unless degrees are explicitly indicated.
- Diagrams are not necessarily drawn to scale.
- The marks for each subpart are shown in brackets.

P1 第1章 · 代数式
英文题目 · 中文解析 · 建议评分细则
本册30题；P1、P2、P3全套25章共750题。

内部复核稿，未完成外部教师逐题审阅。
Original practice material. Not a Pearson examination paper.

1. P1-C01-R01 · P1 · Chapter 1

For $x > 0$, $A = 6x^{3/2}$ and $B = \frac{3}{2}x^{-1/2}$.

(a) Express A/B as kx^n, where k and n are integers. (2)

(b) Given also that $AB = 81$, find

(i) the value of x, (2)

(ii) the exact value of $A + B$. (2)

(Total for Question 1 is 6 marks)

2.

A rectangular card has length $(\sqrt{11} + 2)$ cm and width $(\sqrt{11} - 2)$ cm.

(a) Find the exact area of the card. (2)

(b) A strip of width 1 cm is removed along the full length of the card. Find, in exact form,

(i) the perimeter of the remaining rectangle, (2)

(ii) the fraction of the original area that has been removed. (2)

(Total for Question 2 is 6 marks)

3.

The expression $E = (x + 3)^2 - (x - 1)(x + 5)$ is defined for all real x.

(a) Simplify E. (2)

(b) Given that $E = 3(x + 2)$, find x. (2)

(Total for Question 3 is 4 marks)

4.

Let $F = \dfrac{4x^2 - 25}{2x^2 + 3x - 20}$.

(a) Factorise the numerator and denominator completely. (3)

(b) Simplify F and state all real values of x for which the original expression is undefined. (3)

(Total for Question 4 is 6 marks)

Given that $a = \sqrt{6} + \sqrt{2}$ and $b = \sqrt{6} - \sqrt{2}$,

(a) find the exact values of

(i) $a^2 + b^2$, (2)

(ii) ab. (2)

(b) Hence simplify $\frac{a}{b} + \frac{b}{a}$. (2)

(Total for Question 5 is 6 marks)

For $t > 0$, $V = \dfrac{t^{5/3} + 2t^{2/3}}{t^{1/6}}$.

(a) Express V as the sum of two powers of t. (2)

(b) Given that $t = 9$, find

(i) the exact value of V, (2)

(ii) the exact value of $V/t^{1/2}$. (2)

(Total for Question 6 is 6 marks)

A square has area $(19 - 6\sqrt{10})\ \text{cm}^2$.

(a) Expand $(\sqrt{10} - 3)^2$. (2)

(b) Hence find the exact perimeter of the square. (2)

(Total for Question 7 is 4 marks)

8.

For $x > 0$, $p = \sqrt{x}$ and $q = 1/\sqrt{x}$. It is given that $p - q = 3$.

(a) Show that $p^2 + q^2 = 11$. (3)

(b) Find the exact value of $p + q$. (2)

(Total for Question 8 is 5 marks)

9. P1-C01-R09 · P1 · Chapter 1

A student writes $\sqrt{u+v} = \sqrt{u} + \sqrt{v}$ for positive real numbers u and v.

(a) By squaring the right-hand side, show that this statement is false for every $u > 0$, $v > 0$. (2)

(b) Simplify $\sqrt{48} + \sqrt{75} - \sqrt{27}$. (2)

(Total for Question 9 is 4 marks)

$N = 8^{2/3} \times 16^{-3/4}$.

(a) Find the exact value of N, showing your working. (2)

(b) Express $\dfrac{N}{\sqrt{2}-1}$ in the form $a + b\sqrt{2}$, where a, b are rational. (2)

(Total for Question 10 is 4 marks)

For $x \neq 0$, $E = \dfrac{x^{-2} + 3x^{-1}}{x^{-1}}$.

(a) Simplify E using positive powers of x only. (2)

(b) Find the value of x for which $E = 7$. (2)

(Total for Question 11 is 4 marks)

The dimensions of a box are x cm, $(x + 2)$ cm and $(x - 2)$ cm, where $x > 2$.

(a) Express the volume as a polynomial in x. (2)

(b) Find the total surface area of the box, giving your answer as a polynomial in x. (3)

(Total for Question 12 is 5 marks)

For $x > 0$, $A = \dfrac{\sqrt{50x^3}}{\sqrt{2x}}$.

(a) Show that $A = 5x$. (2)

(b) Given that $B = \sqrt{18x^2}$, express each of the following in simplest form:

(i) B/A, (2)

(ii) $(A + B)(A - B)$. (2)

(Total for Question 13 is 6 marks)

The expression $H = \dfrac{3}{\sqrt{7} - 2} - \dfrac{2}{\sqrt{7} + 2}$.

(a) Express H in the form $a + b\sqrt{7}$, where a, b are rational. (3)

(b) Find the exact value of $3H - \sqrt{7}$. (2)

(Total for Question 14 is 5 marks)

For $y > 0$, $K = \dfrac{(27y^6)^{2/3}}{3y^2}$.

(a) Simplify K. (2)

(b) Given that $K = 12$, find y. (2)

(Total for Question 15 is 4 marks)

A right-angled triangle has perpendicular sides $\sqrt{17} + 1$ cm and $\sqrt{17} - 1$ cm.

(a) Find the exact area of the triangle. (2)

(b) For the same triangle, find, in exact form,

(i) the length of the hypotenuse, (2)

(ii) the perimeter. (2)

(Total for Question 16 is 6 marks)

$f(x) = (2x - 3)(x + 4) - (x - 2)^2$.

(a) Express $f(x)$ in the form $ax^2 + bx + c$. (3)

(b) Find $f(-2)$ and $f(2)$, and hence find $f(2) - f(-2)$. (3)

(Total for Question 17 is 6 marks)

The positive numbers a and b satisfy $a^{1/2} = b^{1/3}$.

(a) Express a in terms of b. (2)

(b) Given also that $b = 64$, find

(i) the exact value of a, (2)

(ii) the exact value of $a^{-3/4}b^{1/2}$. (2)

(Total for Question 18 is 6 marks)

Let $R = \dfrac{x^2 - 6x + 9}{x^2 - 9}$.

(a) Simplify R and state the restrictions on x. (3)

(b) Find the exact value of R when $x = \sqrt{12}$. Give your answer as $a + b\sqrt{3}$. (3)

(Total for Question 19 is 6 marks)

A positive quantity y is related to $x > 0$ by $y = \dfrac{12x^{1/3}}{(3x)^{-2/3}}$.

(a) Express y in the form kx, giving the exact value of k. (3)

(b) Find the exact value of y when $x = 3$. (2)

(Total for Question 20 is 5 marks)

$P = (x - 1)^3 - (x + 1)(x^2 - x + 1)$.

(a) Expand and simplify P. (3)

(b) Find the value of P when $x = -\frac{1}{3}$. (2)

(Total for Question 21 is 5 marks)

$a = 2 + \sqrt{5}$ and $b = 2 - \sqrt{5}$.

(a) Find the exact values of

(i) $a + b$ and ab, (3)

(ii) $a^3 + b^3$, using $a^3 + b^3 = (a + b)^3 - 3ab(a + b)$. (2)

(b) Find $1/a + 1/b$. (2)

(Total for Question 22 is 7 marks)

The algebraic expression $T = 6u^2v - 9uv^2 + 12uv$ is defined for real u, v.

(a) Factorise T completely. (2)

(b) Given that $u = \sqrt{3}$ and $v = 2\sqrt{3}$, find the exact value of T. (3)

(Total for Question 23 is 5 marks)

For $x > 0$, $F = (x^{1/4} + x^{-1/4})^2$.

(a) Expand F as the sum of three terms. (2)

(b) Given that $x = 16$, find

(i) the exact value of F, (2)

(ii) the exact value of $(x^{1/4} - x^{-1/4})^2$. (2)

(Total for Question 24 is 6 marks)

The length L cm of a metal rod is $\dfrac{18}{3+\sqrt{3}}$.

(a) Write L in the form $a + b\sqrt{3}$, where a, b are integers. (2)

(b) The rod is cut into three equal pieces. Find the square of the length of one piece in the form $c + d\sqrt{3}$. (3)

(Total for Question 25 is 5 marks)

$E = (2x + 1)^2 - (x - 2)^2$.

(a) Factorise E completely without first expanding both squares. (2)

(b) Find the coefficient of x in the expansion of E. (2)

(Total for Question 26 is 4 marks)

For $a > 0$ and $b > 0$, $S = \sqrt{a^3 b}\,\sqrt{a b^3}$.

(a) Simplify S. (2)

(b) Given that $ab = 5$, find the exact value of $\dfrac{S + ab}{\sqrt{ab}}$, with a rational denominator. (3)

(Total for Question 27 is 5 marks)

The quantities $U = \dfrac{2x + 6}{x^2 + 5x + 6}$ and $V = \dfrac{1}{x + 2}$ are defined for $x \neq -3, -2$.

(a) Simplify U. (2)

(b) Express $U - V$ as a single fraction and find its value when $x = \sqrt{2}$. Give the value in the form $a + b\sqrt{2}$. (3)

(Total for Question 28 is 5 marks)

29.

A rectangular card has side lengths $(\sqrt{75} + \sqrt{12})$ cm and $(\sqrt{75} - \sqrt{12})$ cm. A square is cut from one corner. The square has the same perimeter as a rectangle of sides $\sqrt{3}$ cm and $2\sqrt{3}$ cm.

(a) Show that the area of the card before cutting is an integer. (2)

(b) For the card remaining after the cut, find

(i) its exact area. (2)

(ii) its exact perimeter, giving your answer in simplest surd form. (2)

(Total for Question 29 is 6 marks)

For $x > 0$, $F(x) = \dfrac{x^{3/2} - 9x^{1/2}}{x^{1/2}}$.

(a) Simplify $F(x)$ and state why $x = 0$ remains excluded. (2)

(b) A positive number a satisfies $F(a^2) = F(a) + 12$.

(i) Find a exactly. (2)

(ii) Hence find $F(1/a)$ exactly. (1)

(Total for Question 30 is 5 marks)

Answers and Suggested Mark Schemes

中文解答与建议评分点

M：方法分；A：准确分，通常依赖对应方法；B：独立得分点。
等价正确解法应获相应分数。评分点为原创建议，并非官方评分细则。

Question 1

P1-C01-R01 · 6 marks

(a) · 2 marks

$4x^{2}$.

系数相除得4；相除时指数相减，$3/2-(-1/2)=2$。

M1	正确处理指数相减
A1	系数与指数均正确

(b) Given also that $AB=81$, find

(i) · 2 marks

$x=9$.

$AB=9x$，故$9x=81$。

M1	化积为9x
A1	得到x=9

(ii) · 2 marks

$\frac{325}{2}$.

$A=162$，$B=1/2$，相加。

M1	将x代回两个原式
A1	得到精确总和

Question 2

P1-C01-R02 · 6 marks

(a) · 2 marks

$7\ \text{cm}^2$.

用平方差：$(\sqrt{11})^2 - 2^2 = 7$。

M1	使用共轭乘积
A1	面积和单位

(b) A strip of width 1 cm is removed along the full length of the card. Find, in exact form,

(i) · 2 marks

$(4\sqrt{11} - 2)$ cm.

新宽为$\sqrt{11} - 3 > 0$；两边相加再乘2。

M1	正确写出新宽
A1	正确周长

(ii) · 2 marks

$\dfrac{\sqrt{11} + 2}{7}$.

去除面积等于原长乘1，再除原面积7。

M1	求去除面积
A1	正确比例

Question 3

P1-C01-R03 · 4 marks

(a) · 2 marks

$2x + 14$.

分别展开为$x^2 + 6x + 9$与$x^2 + 4x - 5$，相减。

M1	正确展开两项
A1	合并同类项

(b) · 2 marks

$x = 8$.

代入(a)得$2x + 14 = 3x + 6$。

M1	使用化简结果建立一次方程
A1	求得8

Question 4

P1-C01-R04 · 6 marks

(a) · 3 marks

$4x^2-25=(2x-5)(2x+5)$; $2x^2+3x-20=(2x-5)(x+4)$.

分子为平方差；分母分解时一次项来自$8x-5x$。

M1	分解分子
M1	分解分母
A1	两式均完全正确

(b) · 3 marks

$F=\dfrac{2x+5}{x+4}$, with $x\neq \frac{5}{2},-4$.

约去共同因子；原分母的两个零点都必须保留为限制，不能因约分而丢失。

M1	正确约分
A1	最简式
B1	保留两个原始限制

Question 5

P1-C01-R05 · 6 marks

(a) find the exact values of

(i) · 2 marks

16.

两个平方分别为$8+4\sqrt{3}$和$8-4\sqrt{3}$；相加根式项抵消。

M1	正确展开至少一个平方
A1	总和16

(ii) · 2 marks

4.

使用平方差$6-2$。

M1	平方差
A1	结果4

(b) · 2 marks

4.

通分为$(a^2+b^2)/(ab)=16/4$。

M1	正确通分并引用(a)
A1	结果4

Question 6

P1-C01-R06 · 6 marks

(a) · 2 marks

$t^{3/2} + 2t^{1/2}$.

两项分别减去指数$1/6$。

M1	对两项都减指数
A1	两个幂正确

(b) Given that $t = 9$, find

(i) · 2 marks

33.

$9^{1/2} = 3,\ 9^{3/2} = 27$。

M1	正确计算分数指数
A1	结果33

(ii) · 2 marks

11.

由(a)可得$V/t^{1/2} = t + 2,$ 代入9。

M1	化商为t+2
A1	结果11

Question 7

P1-C01-R07 · 4 marks

(a) · 2 marks

$19-6\sqrt{10}$.

平方含$10+9$及交叉项$-6\sqrt{10}$。

M1	写出交叉项
A1	正确展开

(b) · 2 marks

$4(\sqrt{10}-3)$ cm.

$\sqrt{10}>3$，故正边长为$\sqrt{10}-3$；周长为边长四倍。

M1	选取正边长
A1	正确周长及单位

Question 8

P1-C01-R08 · 5 marks

(a) · 3 marks

$p^2+q^2=11.$

$pq=1$，故$(p-q)^2=p^2+q^2-2=9$。

M1	识别pq=1
M1	展开差的平方
A1	推得11

(b) · 2 marks

$\sqrt{13}.$

$(p+q)^2=11+2=13$；p、q均为正，所以取正平方根。

M1	求和的平方
A1	选正根

Question 9

P1-C01-R09 · 4 marks

(a) · 2 marks

$(\sqrt{u}+\sqrt{v})^2 = u+v+2\sqrt{uv} > u+v.$

交叉项严格为正，两个非负数的平方不同，因此原数不同。

M1	正确展开平方
A1	用正交叉项得到严格结论

(b) · 2 marks

$6\sqrt{3}.$

各项提出完全平方因子，系数为4、5、3。

M1	将三项化为同类根式
A1	合并正确

Question 10

P1-C01-R10 · 4 marks

(a) · 2 marks

$\frac{1}{2}$.

$8^{2/3}=4$；$16^{-3/4}=1/8$。

M1	正确处理两个分数指数
A1	结果1/2

(b) · 2 marks

$\frac{1}{2}+\frac{1}{2}\sqrt{2}$.

代入N后，分子分母同乘$\sqrt{2}+1$；分母的共轭乘积为1。

M1	乘共轭式
A1	得到所要求形式

Question 11

P1-C01-R11 · 4 marks

(a) · 2 marks

$E = \dfrac{1}{x} + 3$.

逐项除以x^{-1}，指数相减。

M1	分别处理两项
A1	正确正指数形式

(b) · 2 marks

$x = \frac{1}{4}$.

$1/x = 4$，两边取倒数，所得值满足$x \neq 0$。

M1	使用化简式
A1	求得$1/4$

Question 12

P1-C01-R12 · 5 marks

(a) · 2 marks

$V = x^3 - 4x \text{ cm}^3$.

先将$(x+2)(x-2)$写成x^2-4再乘x。

M1	利用平方差
A1	正确多项式及单位

(b) · 3 marks

$(6x^2 - 8) \text{ cm}^2$.

三对面的面积和为$2[x(x+2)+x(x-2)+(x+2)(x-2)]$，展开并合并。

M1	正确计入三对面
M1	展开三个乘积
A1	多项式及单位

Question 13

P1-C01-R13 · 6 marks

(a) · 2 marks

$A = \sqrt{25x^2} = 5x.$

根号内的商为$25x^2$；由x正可取5x。

M1	合并商并化简根号内
A1	正确使用x为正

(b) Given that $B = \sqrt{18x^2}$, express each of the following in simplest form:

(i) · 2 marks

$\dfrac{3\sqrt{2}}{5}.$

$B = 3x\sqrt{2}$，与5x相除。

M1	化简B
A1	约去x

(ii) · 2 marks

$7x^2.$

用$A^2 - B^2 = 25x^2 - 18x^2$。

M1	使用平方差
A1	结果7x平方

Question 14

P1-C01-R14 · 5 marks

(a) · 3 marks

$\frac{10}{3}+\frac{1}{3}\sqrt{7}$.

两个分母的共轭乘积均为3；通分后分子$3(\sqrt{7}+2)-2(\sqrt{7}-2)=\sqrt{7}+10$。

M1	两个分式有理化
M1	正确相减分子
A1	所要求形式

(b) · 2 marks

10.

代入(a)，根式部分相消。

M1	正确代入化简式
A1	结果10

Question 15

P1-C01-R15 · 4 marks

(a) · 2 marks

$3y^2$.

分子为$9y^4$，除以$3y^2$。

M1	正确计算分子
A1	最简式

(b) · 2 marks

$y = 2$.

$y^2 = 4$，利用y正只保留2。

M1	解平方关系
A1	使用正值限制

Question 16

P1-C01-R16 · 6 marks

(a) · 2 marks

$8\ \text{cm}^2$.

面积为两直角边乘积的一半，即$(17-1)/2$。

M1	正确面积式
A1	结果8及单位

(b) For the same triangle, find, in exact form,

(i) · 2 marks

$6\ \text{cm}$.

两直角边平方和为$2(17+1)=36$。

M1	使用勾股并消去交叉项
A1	正根6

(ii) · 2 marks

$(2\sqrt{17}+6)\ \text{cm}$.

直角边之和为$2\sqrt{17}$，再加斜边6。

M1	正确合并边长
A1	周长表达式

Question 17

P1-C01-R17 · 6 marks

(a) · 3 marks

$f(x)=x^2+9x-16.$

首项乘积为$2x^2+5x-12$；减去x^2-4x+4。

M1	展开乘积和平方
M1	正确处理减号
A1	三个系数

(b) · 3 marks

$f(-2)=-30$, $f(2)=6$; difference 36.

代入得到$4-18-16$与$4+18-16$；按所给顺序相减。

M1	两次正确代入
A1	两函数值
A1	差为36

Question 18

P1-C01-R18 · 6 marks

(a) · 2 marks

$a = b^{2/3}$.

两边平方。

M1	使用指数幂运算
A1	正确指数

(b) Given also that $b = 64$, find

(i) · 2 marks

16.

64的立方根为4，再平方。

M1	正确开立方
A1	得到16

(ii) · 2 marks

1.

$16^{-3/4} = 1/8$，$64^{1/2} = 8$。

M1	正确负分数指数
A1	结果1

Question 19

P1-C01-R19 · 6 marks

(a) · 3 marks

$R = \dfrac{x-3}{x+3};\ x \neq \pm 3.$

分子$(x-3)^2$，分母$(x-3)(x+3)$；约分后保留原限制。

M1	分解并约分
A1	最简式
B1	两个限制

(b) · 3 marks

$7-4\sqrt{3}.$

$x=2\sqrt{3}$。将$(2\sqrt{3}-3)/(2\sqrt{3}+3)$有理化，得$(21-12\sqrt{3})/3$。

M1	正确代入并乘共轭
M1	展开平方及分母
A1	所要求形式

Question 20

P1-C01-R20 · 5 marks

(a) · 3 marks

$y = 12\,3^{2/3}x$.

除以负指数幂等于乘以正指数幂；x的指数相加为1。

M1	将除法改为乘法
M1	合并x的指数
A1	精确系数

(b) · 2 marks

$36\sqrt[3]{9}$.

代入得到$36\,3^{2/3}$，即所给根式。

M1	正确代入
A1	精确值

Question 21

P1-C01-R21 · 5 marks

(a) · 3 marks

$-3x^2+3x-2.$

第二个乘积为x^3+1，首项立方为x^3-3x^2+3x-1。

M1	展开立方
M1	展开第二乘积
A1	合并正确

(b) · 2 marks

$-\frac{10}{3}.$

代入得$-1/3-1-2$。

M1	正确处理负数平方
A1	精确结果

Question 22

P1-C01-R22 · 7 marks

(a) Find the exact values of

(i) · 3 marks

$a+b=4,\ ab=-1$.

和中根式相消；积用平方差$4-5$。

B1	和为4
M1	计算共轭乘积
A1	积为负1

(ii) · 2 marks

76.

使用题目给出的恒等式，得$64-3(-1)(4)=76$。

M1	正确代入三个已知量
A1	结果76

(b) · 2 marks

-4.

通分为$(a+b)/(ab)=4/(-1)$。

M1	通分
A1	结果负4

Question 23

P1-C01-R23 · 5 marks

(a) · 2 marks

$3uv(2u-3v+4)$.

三项最大公因式为3uv；依次除去得到括号中的三项。

M1	提取公因式3uv
A1	括号正确

(b) · 3 marks

$72-72\sqrt{3}$.

$3uv=18$，括号为$4-4\sqrt{3}$，相乘。

M1	代入化简后的两因子
M1	正确处理根式乘积
A1	精确结果

Question 24

P1-C01-R24 · 6 marks

(a) · 2 marks

$x^{1/2}+2+x^{-1/2}$.

交叉项为$2x^{0}=2$；两个平方的指数分别为$1/2$与$-1/2$。

M1	正确交叉项
A1	两平方项

(b) Given that $x=16$, find

(i) · 2 marks

$\frac{25}{4}$.

代入得到$4+2+1/4$。

M1	正确计算正负半次幂
A1	精确值

(ii) · 2 marks

$\frac{9}{4}$.

只改变交叉项的符号：$4-2+1/4=9/4$。

M1	使用差的平方
A1	结果$9/4$

Question 25

P1-C01-R25 · 5 marks

(a) · 2 marks

$L = 9 - 3\sqrt{3}$.

乘共轭后分母为6，分子为$18(3-\sqrt{3})$。

M1	使用共轭有理化
A1	正确整数系数

(b) · 3 marks

$(12 - 6\sqrt{3})\ \text{cm}^2$.

一段长$3-\sqrt{3}$；平方为$9+3-6\sqrt{3}$。

M1	先将全长除以3
M1	正确展开平方
A1	精确式及单位

Question 26

P1-C01-R26 · 4 marks

(a) · 2 marks

$E = (x+3)(3x-1)$.

平方差得$[(2x+1)-(x-2)][(2x+1)+(x-2)] = (x+3)(3x-1)$。

M1	使用平方差
A1	正确两因子

(b) · 2 marks

8.

直接展开得$3x^2+8x-3$，一次项系数8。

M1	正确展开或合并交叉项
A1	系数8

Question 27

P1-C01-R27 · 5 marks

(a) · 2 marks

a^2b^2.

合并根式得$\sqrt{a^4b^4}$，取非负平方根。

M1	合并根式
A1	正确平方根

(b) · 3 marks

$6\sqrt{5}$.

先得S=25，式子为$30/\sqrt{5}$；有理化得$6\sqrt{5}$。

M1	用ab求S
M1	分母有理化
A1	最简根式

Question 28

P1-C01-R28 · 5 marks

(a) · 2 marks

$U = \dfrac{2}{x+2}$.

分子$2(x+3)$，分母$(x+2)(x+3)$，约去共同因子。

M1	正确因式分解
A1	最简式

(b) · 3 marks

$U - V = \dfrac{1}{x+2}$; value $1 - \frac{1}{2}\sqrt{2}$.

先同分母相减；$1/(2+\sqrt{2}) = (2-\sqrt{2})/2$。

M1	正确单分式
M1	有理化
A1	指定形式

Question 29

P1-C01-R29 · 6 marks

(a) · 2 marks

$63\ \text{cm}^2$.

利用平方差：$(\sqrt{75}+\sqrt{12})(\sqrt{75}-\sqrt{12})=75-12=63$。

M1	平方差
A1	面积63

(b) For the card remaining after the cut, find

(i) · 2 marks

$225/4\ \text{cm}^2$.

小矩形周长为$6\sqrt{3}$，故正方形边长$3\sqrt{3}/2$，剪去面积$27/4$；剩余$63-27/4=225/4$。

M1	由周长求正方形边长
A1	剩余面积

(ii) · 2 marks

$20\sqrt{3}\ \text{cm}$.

从角上剪去正方形时，失去的两段外边各以等长切边补回，周长不变；$2(2\sqrt{75})=20\sqrt{3}$。

B1	说明周长不变
A1	最简根式

Question 30

P1-C01-R30 · 5 marks

(a) · 2 marks

$F(x) = x - 9,\ x > 0.$

分子为$\sqrt{x}(x-9)$，约去$\sqrt{x}$；原式在$x=0$分母为零，约分不能扩大定义域。

M1	提取并约去公因式
B1	原式定义域

(b) A positive number a satisfies $F(a^2) = F(a) + 12$.

(i) · 2 marks

$a = 4.$

$a^2 - 9 = a - 9 + 12$，即$(a-4)(a+3)=0$；由$a>0$取4。

M1	建立二次方程
A1	合法根

(ii) · 1 marks

$-35/4.$

代入已化简的式子：$F(1/4) = 1/4 - 9 = -35/4$。

B1	准确值